

Date	03 July 2026
Team ID	SWTID-2026-9440
Team Size	4
Team Leader	Bodanapu Sai Krishna Reddy
Team Member 1	Anand Ch
Team Member 2	Singam Hemasree
Team Member 3	Damavarapu Sanjana
Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Project Planning

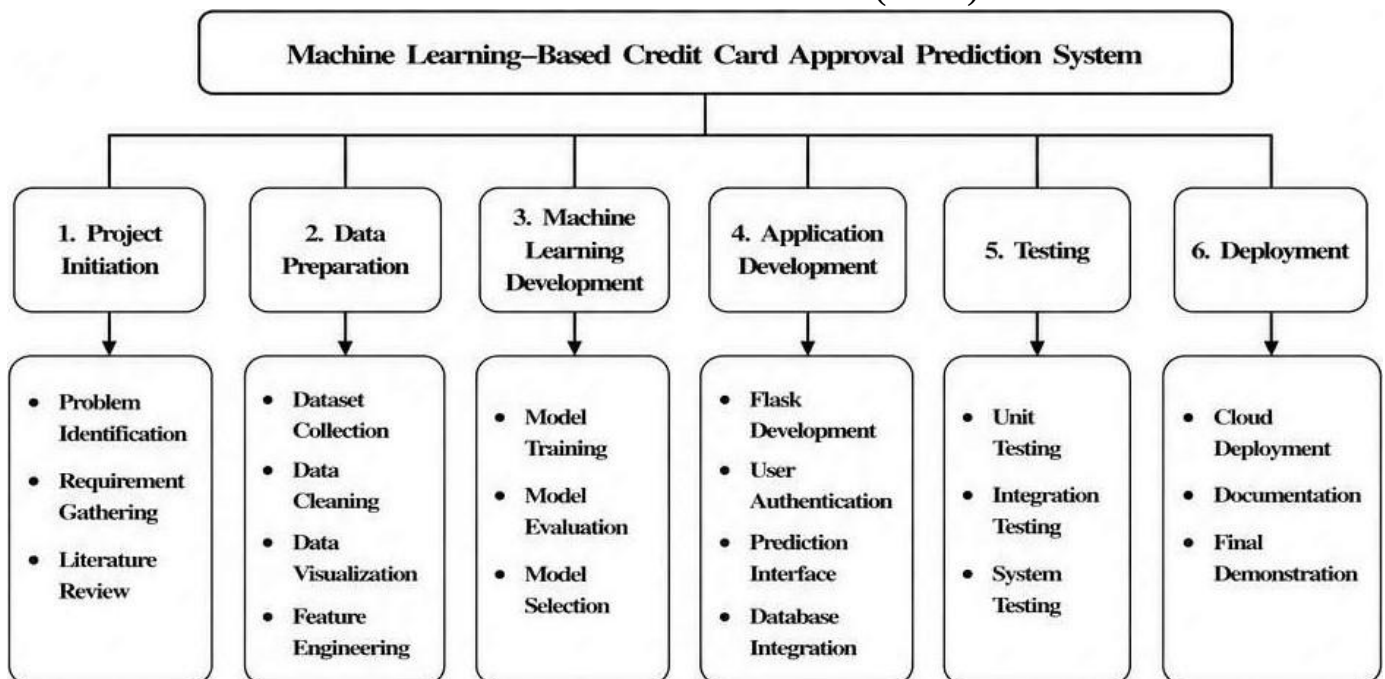
Introduction

The Project Planning document defines the overall execution strategy for developing the Machine Learning—Based Credit Card Approval Prediction System. It identifies project phases, major activities, deliverables, team responsibilities, and the planned development sequence required for successful project completion.

Project Objectives

- Develop a secure Machine Learning—based credit card approval prediction system.
- Automate applicant eligibility evaluation.
- Achieve high prediction accuracy using supervised learning algorithms.
- Build a responsive Flask web application.
- Store applicant and prediction data securely.
- Deploy a scalable solution suitable for real-world banking applications.

Work Breakdown Structure (WBS)



Conclusion

The Work Breakdown Structure provides a systematic approach for executing the project by dividing the complete development process into manageable phases, improving planning, coordination, and project monitoring.

Pmject Schedule and Resource Planning

The following schedule outlines the planned development activities, estimated duration, project milestones, resource allocation, and potential project risks associated with the Machine Learning-Based Credit Card Approval Prediction System. The plan ensures systematic execution, efficient resource utilization, timely completion, and successful deployment of the proposed solution.

1. Project Timeline

		Estimated Duration	Deliverable
1	Requirement Analysis	1 Week	Software Requirement Specification (SRS)
2	Dataset Collection and Analysis	1 week	Prepared and Cleaned Dataset
3	Data Preprocessing and Feature Engineering	1 Week	Cleaned and Processed Dataset
4	Machine Learning Model Development	2 Weeks	Trained and Evaluated Prediction Models
5	Flask Web Application Development	2 Weeks	Functional Web Application
6	System Testing and Validation	week	Testing and Validation
7	Deployment and Documentation	1 week	Deployed Application and final Project

2. Milestones

Milestone	Expected Outcome
Requirement Analysis Completed	Approved Requirements
Dataset	Clean Dataset Ready for Training
Learning Model Completed	Best prediction Model Selected
Web Application Developed	Prediction System Successfully Integrated
System Testing Completed	Verified Functional System
Deployment Completed	Application Successfully Hosted and Ready for Demonstration

3. Risk Assessment

	Mitigation Strategy
Low Data Quality	Perform thorough validation, and data cleaning.
Prediction Accuracy	Train multiple machine learning algorithms and test performance.
Deployment Failure	Test deployment thoroughly using Docker containers before production deployment on server.
Security Vulnerabilities	Implement secure authentication, encrypted communication, and protected database access.
Project Schedule	Monitor milestones regularly and distribute development tasks efficiently among team members.

Resource Allocation

4. Resource Allocation

Resource	Python	Machine Learning Development and Backend Programming.
Backend Framework	Python 3.x	Develops REST API, business logic, authentication, and prediction services.
	JavaScript, CSS3, Bootstrap 5	Builds a responsive and user-friendly Web interface.
	MySQL	Stores applicant information, accounts, application and prediction history securely
Machine Learning Libraries	Scikit-learn, pandas, NumPy	Supports data preprocessing, feature engineering, model training, evaluation, and prediction.
Development Tools	Visual Studio Code, Jupyter, Git, Docker, Flask	Development, machine learning, and API testing application framework for experimentation.
Version Control	Git and GitHub	Manages project version tracking, and maintenance.
Cloud		Hosts the Flask web application with reliable cloud infrastructure and scalable

Containerization		Packages the application into portable containets for consistent testing, and deploynw•nt environments.
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